

Data Science Bootcamp Week-7 Assignment

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Q-1) How do you assess the statistical significance of an insight?

Answer:

To assess whether an insight is statistically significant, I typically perform hypothesis testing. This involves setting up a null hypothesis — usually something like “there is no difference” or “no effect.” I then calculate a p-value using an appropriate test (like a t-test or chi-squared test) to measure the probability that the observed results could have happened by chance. If the p-value is below a commonly accepted threshold (usually 0.05), we reject the null hypothesis and say the insight is statistically significant. Essentially, it means we're confident the result isn't just random noise.

Q-2) What is the Central Limit Theorem? Explain it. Why is it important?

Answer:

The Central Limit Theorem (CLT) is one of the foundational concepts in statistics. It states that, regardless of the original distribution of a dataset, the distribution of the sample means will tend to be normal if the sample size is large enough. This is really important because it allows us to make inferences about populations from sample data — even when the original data isn't normally distributed. Thanks to the CLT, we can apply many statistical techniques that assume normality, like confidence intervals and z-tests, with more confidence.

Q-3) What is the statistical power?

Answer:

Statistical power is the probability that a test will correctly detect a true effect — in other words, the likelihood that we reject a false null hypothesis. A powerful test reduces the risk of a Type II error, which is when we fail to detect an effect that actually exists. Power increases with a larger sample size, a more noticeable effect size, and a lower level of noise or variability in the data. When planning experiments, it's important to ensure we have enough power to detect meaningful differences.

Q-4) How do you control for biases?

Answer:

Controlling for biases is key to making sure our insights are valid and trustworthy. I usually start by being careful with data collection — ensuring it's as random and representative as

possible to avoid selection bias. In experiments, I'd use randomization to assign participants to groups and sometimes blinding to reduce influence. I also look out for potential outliers or anomalies that could skew the data, and make sure to include control groups for comparison. Statistical techniques like regression can also help adjust for known biases and confounding variables.

Q-5) What are confounding variables?

Answer:

Confounding variables are factors that can mess up the relationship you're trying to study. They're variables that affect both the independent and dependent variables, which can lead to misleading conclusions. For example, if you're studying whether coffee causes heart disease, a confounding variable could be stress — since it can lead to both increased coffee consumption and heart issues. To handle confounders, we can use statistical controls, stratify our data, or design experiments in ways that isolate the variables we care about.

Q-6) What is A/B testing?

Answer:

A/B testing is a straightforward but powerful experiment used to compare two versions of something — like two website designs or app features — to see which one performs better. You split users randomly into two groups: one sees version A (the control), and the other sees version B (the variation). You then measure the outcome you care about, like click-through rate or conversion, and use statistical analysis to determine whether the difference in performance is significant or just due to chance. It's commonly used in product and marketing optimization.

Q-7) What are confidence intervals?

Answer:

Confidence intervals give us a range of values that we believe likely contains the true value we're estimating — like a population mean. For example, a 95% confidence interval means that if we repeated our sampling process many times, about 95% of those intervals would include the actual population value. They give a sense of how precise our estimate is: narrower intervals suggest more confidence in our estimate, while wider ones indicate more uncertainty. It's a great way to quantify the reliability of our data insights.